

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1.A media company wants to migrate its video streaming platform to the cloud.The system must handle millions of concurrent users with minimal buffering.Design a cloud architecture using scalability and caching techniques.Explain how load balancing and auto-scaling can ensure smooth streaming.Discuss cost optimization strategies for such high-demand workloads.

Understanding the Scenario

In this case, a media company wants to move its video streaming platform to the cloud. Since millions of users may watch videos at the same time, the biggest challenge is ensuring smooth streaming without buffering or service interruptions. The system should also be able to handle sudden increases in traffic during live events or popular content releases.

Risks and Issues Identified

When many users access the platform the servers may get overloaded.Users msy experience low video quality and buffering.While streaming videos a large amount of bandwidth will get consumed.Problem in a single server may affect all the other servers.

Load balancing and Auto scaling:

To over come these issues load balancing is used as it prevents any single server from receiving too many requests this improves the performance and r5reduces the chance of failures.

The number users can always vary depending on the event if it is a major sporting even or a movie the users will significantly increase and to manage tht we use Auto scaling it removes unnecessary servers when traffic decreases.

And inn the end the user will recieve consistent streaming and the company avoids paying for unused resources.

Decision Justification

CDN is selected, load balancing, and auto-scaling because they directly address the main challenges of a video streaming platform. The CDN reduces buffering, load balancing improves availability, and auto-scaling ensures the system can handle millions of users without performance degradation.

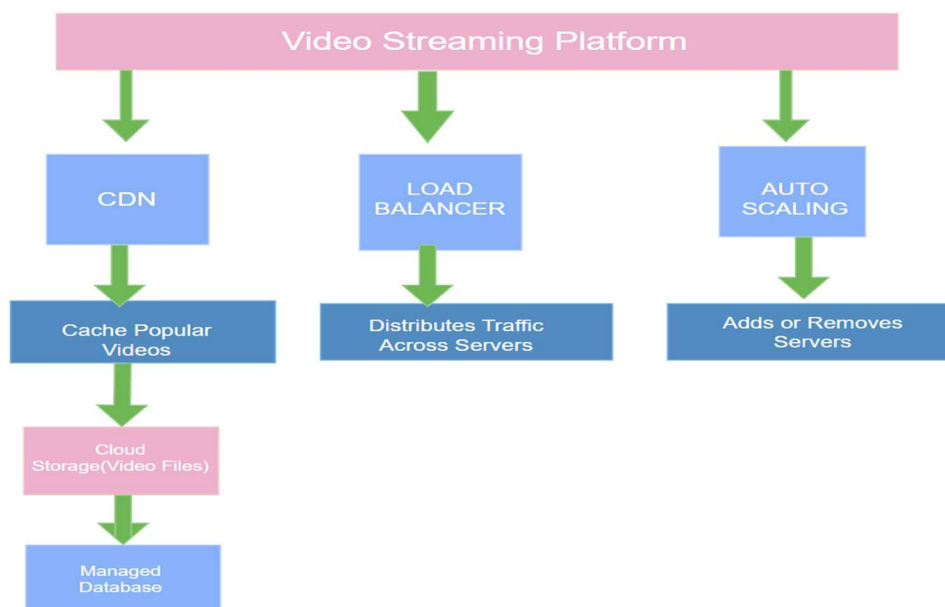


Fig1:Video Streaming Platform Architecture

2.A fintech startup is deploying applications using containers across multiple cloud providers.They face challenges in maintaining consistency and managing deployments.Propose a solution using container orchestration and multi-cloud strategies.Explain how service discovery and scaling can be handled efficiently.Highlight the benefits and risks of a multi-cloud deployment model.

Understanding the Scenario

A fintech startup is deploying its applications using containers across multiple cloud providers. Since financial applications handle sensitive transactions and customer data, the deployment must be consistent, secure, and highly available. The company is facing difficulties in managing applications across different cloud environments.

Risks and Issues Identified

The main challenges in this scenario are:

- Maintaining the same configuration across multiple cloud providers.
- Managing large numbers of containers efficiently.
- Ensuring services can communicate with each other without failures.
- Avoiding downtime during updates and deployments.
- Increased complexity in monitoring and security management.

Proposed Solution

To solve these challenges, I would use Docker containers along with Kubernetes for container orchestration.

Applications would be packaged into Docker containers and deployed on Kubernetes clusters running across different cloud providers such as AWS, Azure, and Google Cloud.

A CI/CD pipeline can automatically build, test, and deploy applications whenever changes are made. This ensures consistency across all environments and reduces manual errors.

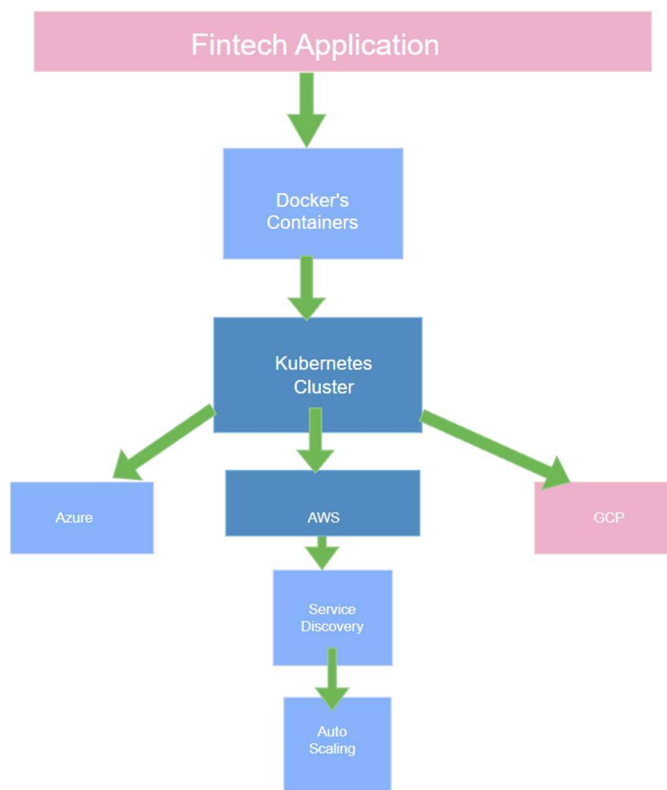
How Service Discovery and Scaling Can Be Handled

Kubernetes provides built-in service discovery using DNS. Instead of communicating through IP addresses, services can communicate using service names. This makes communication more reliable even when containers are created or removed.

For scaling, Kubernetes can automatically increase or decrease the number of container instances based on CPU usage, memory usage, or incoming traffic. This helps the application handle varying workloads without manual intervention.

Decision Justification

I selected Kubernetes because it provides a common platform for managing containers regardless of the cloud provider. This simplifies deployment, improves scalability, and ensures application consistency across all environments.



3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Understanding the Scenario

An e-learning platform has experienced data loss due to accidental deletion in the cloud. Since the platform stores student records, course materials, assignments, and examination data, recovering lost information quickly is extremely important.

Risks and Issues Identified

The major risks include:

- Permanent loss of important educational data.
- Service disruption affecting students and faculty.
- Loss of examination records and learning progress.
- Reduced trust among users.

Proposed Solution

I would implement a cloud-native backup and disaster recovery strategy consisting of:

- Automated backups.
- Cross-region replication.
- Object versioning.
- Database snapshots.

All critical data would be backed up automatically and replicated to a secondary region. Object versioning would be enabled so that deleted files can be restored easily.

Database snapshots would be taken periodically to provide additional recovery options.

Achieving RTO and RPO

Recovery Time Objective (RTO)

RTO refers to how quickly the system can be restored after a failure.

To achieve a low RTO:

- Maintain a standby environment in another region.
- Use automated failover mechanisms.
- Store backup configurations along with application data.
 1. Enable automatic backups for all critical resources.
 2. Store backup copies in a different region.
 3. Regularly test backup restoration procedures.
 4. Create a disaster recovery plan and document all recovery steps.
 5. Train administrators to handle emergency situations.

Decision Justification

The combination of backups, replication, and versioning ensures that accidental deletions or system failures do not result in permanent data loss. Recovery becomes faster and more reliable.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Understanding the Scenario

A government agency plans to adopt a hybrid cloud model. Sensitive information must remain on-premises due to regulatory requirements, while non-sensitive applications can be moved to the public cloud to improve scalability and efficiency.

Risks and Issues Identified

The main concerns are:

- Protection of confidential government data.
- Compliance with legal and regulatory requirements.
- Secure communication between on-premises and cloud environments.
- Complexity in managing different infrastructures.

Proposed Hybrid Architecture

I would design a hybrid architecture where:

- Sensitive databases and critical applications remain in the on-premises data center.
- Public-facing applications and analytics workloads run in the public cloud.
- Both environments are connected through a secure VPN or dedicated private connection.

This allows the organization to maintain control over sensitive data while benefiting from cloud scalability.

5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Understanding the Scenario

A SaaS provider must ensure that its application remains available even if an entire cloud region experiences an outage. Customers expect uninterrupted service, so high availability and fault tolerance are critical requirements.

Risks and Issues Identified

The main risks found here is that it can face regional cloud outages, Service downtime will affect the customers, When the server goes through a failure the data will be lost, If the system has any issues the detection could be delayed.

Multi-Deployment:

To overcome these issues and risks we deploy the same application across at least two geographically separate cloud regions.

We can take two regions primary and secondary region,

The primary region will handle all the normal operations and meanwhile the secondary region will remain ready to take over if a failure occurs.

Monitoring and Alerting Strategy: Continuous monitoring will be allotted for:

- CPU utilization.
- Memory usage.
- Network latency.
- Database performance.
- Error rates.

Alerts should be sent through email, SMS, or incident management platforms whenever abnormal activity is detected.

Decision Justification

A multi-region architecture eliminates the single point of failure problem. Even if one region becomes unavailable, another region can continue serving users, ensuring business continuity.